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课题研究报告 · RESEARCH STUDY

前线部署工程师：AI“最后一公里”的组织机制

The Forward-Deployed Engineer: An Organizational Mechanism for AI's Last Mile

关于楔子选择、残留物产品化与清晰产权的课题研究

A research study on wedge selection, productized residue and clear title

研究底本 SOURCE
Cuofano, G. The Forward-Deployed Engineer (2026)

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摘要 *Abstract*

前沿模型日益成为可租用的商品，但大多数企业AI项目仍停滞在演示与损益表之间。本研究以Gennaro Cuofano所著《前线部署工程师》（2026）为底本，对FDE这一角色进行结构化的分析性阅读，探讨它如何弥合通用能力与特定生产价值之间的“最后一公里”，以及其价值如何在供应商、买方与个人之间分配。研究构建“问题—机制—治理”三层框架，并以底本声明的四项可证伪模式为工作假说。主要发现有四：**第一**，楔子选择是治理决策，实质是对“信任时钟”的预算管理；**第二**，嵌入是萃取特定性的方法，构建的产物还包括客户基于证据的信心；**第三**，模式的经济性来自级联与收获两次转化，二者只在被排期保护时发生；**第四**，嵌入式工程师既是最快的能力转移也是最深的依赖机制，差别由合同结构决定。开创企业案例与保险业综合案例提供了方向性支持与一致性检验，但四项假说尚未在部署总体上得到检验。文末讨论三方启示、六类失败模式、研究局限与研究议程。

Frontier models are increasingly a rentable commodity, yet most enterprise AI initiatives still stall between the demo and the P&L. Taking Gennaro Cuofano's The Forward-Deployed Engineer (2026) as its source volume, this study offers a structured analytical reading of the FDE role: how it closes the last mile between general capability and specific production value, and how that value is distributed among vendor, buyer and individual. It builds a three-level framework of problem, motion and governance, and adopts the source's four falsifiable patterns as working hypotheses. Four findings emerge. **First**, wedge selection is a governance decision that budgets the trust clock. **Second**, embedding is a method for extracting the specific, and the build produces the customer's evidence-based confidence as well as machinery. **Third**, the motion's economics come from two conversions, cascade and harvest, which happen only when scheduled. **Fourth**, the embedded engineer is both the fastest capability transfer and the deepest dependency mechanism, and contract structure decides which. The originator's public record and an insurance composite give directional support and a consistency check, but no hypothesis has been tested on a population of deployments. The study closes with implications, six failure families, limitations and a research agenda.

关键词 前线部署工程师；最后一公里；楔子选择；残留物产品化；清晰产权；企业AI落地

KEYWORDS forward-deployed engineer; last mile; wedge selection; productized residue; clear title; enterprise AI deployment

研究问题 *Research Questions*

RQ1	FDE以何种机制把通用能力转化为企业的特定生产价值?	By what mechanisms does the FDE turn general capability into specific production value?
RQ2	哪些设计选择决定部署能否在信任窗口内见效?	Which design choices decide whether a deployment succeeds within the trust window?
RQ3	复利价值如何积累，又由什么决定其归属?	How does compounding value accumulate, and what decides who keeps it?
RQ4	底本的四项承重模式得到多大支持，又需怎样检验?	How well supported are the source's four load-bearing patterns, and how should they be tested?

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第 1 章 · CHAPTER 1

01 研究背景与问题提出

Background and Problem Statement

前沿模型已可按需租用，但多数企业AI项目仍止步于演示与损益表之间。本章界定这一"最后一公里"问题，并说明本研究的出发点。

Frontier models can be rented on demand, yet most enterprise AI initiatives still stall between the demo and the P&L. This chapter frames that last-mile problem and the study's point of departure.

中文　　ENGLISH

1.1 能力商品化与价值落地之间的落差

Commoditized Capability, Undelivered Value

底本从当下周期中日益清晰的一个悖论出发。前沿能力已呈现商品属性：任何企业都可以通过API，在短时间内租用与竞争对手相同的模型。然而，少数企业却把这一共享投入转化为"看似服务、实为软件"的利润率，以及价值数十亿美元、按结果计价的合同；与此同时，大多数企业AI项目仍在演示与损益表之间夭折。底本认为，差异不在模型，而在**最后一公里**——即模型所能做的，与企业真实的工作、数据和人员在生产环境中允许它做的，二者之间的距离。

The source volume opens from a paradox that the present cycle has made visible. Frontier capability behaves like a commodity: any enterprise can rent the same models as its competitors, by API, at short notice. Yet a small set of firms converts this shared input into software-class margins on work that resembles services, and into outcome-based contracts worth billions, while the median enterprise initiative dies between the demo and the profit-and-loss statement. The volume locates the difference not in the model but in the **last mile**: the distance between what a model can do and what an organization's actual work, data and people will allow it to do in production.

前线部署工程师（forward-deployed engineer，FDE）正是这个时代用以弥合这一距离的角色：工程师嵌入客户的约束所在，用客户的真实案例、在客户的真实系统中构建生产级机制，同时肩负一项指向产品的"第二职责"。底本认为，AI使这一早已存在的角色成为应用智能的主导市场进入方式；技术可得性与组织吸收能力之间的不对称，正是本研究的出发点。

The forward-deployed engineer (FDE) is the role the era adopted to close that distance: an engineer embedded at the customer's constraint, building production machinery from the customer's real cases inside the customer's real systems, while carrying a second duty back toward the product. The source argues that AI made this older role the dominant go-to-market for applied intelligence; the asymmetry between technical availability and organizational absorption is the starting point of this study.

1.2 三种先行方案及其结构性缺口

Three Earlier Answers and Their Structural Gaps

底本把FDE视为对一个老问题的第四种回答：复杂技术如何从供应商手中跨入客户的运营？解决方案工程师隶属销售，负责配置产品，签约即离场，它弥合的是交易而非差距。咨询项目是一座按时间计价的杠杆金字塔：交付的是分析而非安装，激励与能力转移相悖，也无法承担生产责任，因为它从不构建产生结果的机制。随后，数据平台一代发现，其通用产品若没有工程师在现场构建"特定部

The volume reads the FDE as the fourth answer to an old question: how does complex technology cross from a vendor's hands into a customer's operations? The solutions engineer, attached to sales, configured the product and left at signature; it closed deals, not gaps. The consulting engagement, a leverage pyramid priced by time, delivered analysis rather than installation, was paid against transfer, and could not hold production accountability because it never built the machinery

分”便根本无法售出——这一模式起初被讥为不可扩展的服务业务，后来却在公开财报中得到证明。

that would produce the outcome. The data-platform generation then discovered that its general product could not be sold at all without engineers building the specific on site, a motion first derided as unscalable services and later vindicated in public filings.

表 1-1 四种技术落地方案的比较
Table 1-1 *Four answers to the crossing problem*

方案 Answer	交付物 Deliverable	对能力转移的激励 Incentive on transfer	生产责任 Accountability
解决方案工程师 Solutions engineer	成功的演示 A successful demo	签约后即离场 Exits at signature	无 None
咨询项目 Consulting engagement	分析与汇报材料 Analysis and the deck	与转移相悖（按时计费） Against transfer (billed by time)	结构上无法承担 Structurally unable
平台型现场工程 Platform field engineering	由通用平台构建的特定系统 Specific system built from a general platform	经验回流产品 Learning flows back to product	对部署负责 Owns the deployment
AI时代的FDE AI-era FDE	通过冻结评测集的生产单元 Production unit passing a frozen suite	转移是收尾交付物 Transfer is the closing deliverable	对客户结果负责 Accountable for the outcome

来源 Source: 本研究据 Cuofano (2026) 第一、六章整理 / compiled from Chapters I and VI.

1.3 为何是现在：四股力量
Why Now: Four Forces

底本列出使当前周期成为这一模式关键时刻的四股力量。其一，上游商品化：前沿模型趋同，竞争焦点转向“安装”，而安装正是该模式的产出。其二，下游停滞：AI承诺巨大而部署滞后，形成已获资金支持的需求。其三，横向模仿：实验室、云厂商、数据平台与服务公司都已派驻FDE。其四，窗口期：FDE在这十年手工完成的整理工作正被编码进产品，手工形态自有其保质期——对积累残留物的供应商意味着紧迫，对观望的买方则意味着代价。

The source identifies four forces that make the present cycle the motion's moment. Commoditization above: as frontier models converge, the competitive question moves to the installation, which is exactly what the motion manufactures. Stall below: large AI commitments with lagging deployment create funded demand. Imitation across: labs, clouds, data platforms and services firms all now field FDEs. And the window: the manual curation FDEs perform this decade is being encoded into product, so the motion's manual form has a shelf life, which is urgent for vendors accumulating residue and costly for buyers who wait.

核心洞见 KEY INSIGHT

本研究的核心命题 *The study's central proposition*

FDE并非服务业务的权宜之计，而是一种**组织机制**：它把通用能力转化为特定的生产价值，并把跨客户重复出现的部分沉淀为产品。其成败取决于结构设计，而非个人英雄主义。

The FDE is not a stopgap services business but an **organizational mechanism**: it converts general capability into specific production value and condenses what repeats across customers into product. Its success depends on structural design, not on individual heroics.

第 2 章 · CHAPTER 2

02 核心概念与理论框架

Core Concepts and Analytical Framework

本章确立FDE的操作性定义，梳理角色演进的五个"年代"，并提出贯穿全文的三层分析框架。

This chapter fixes an operational definition of the FDE, traces five vintages of the role, and sets out the three-level framework used throughout the study.

中文　　ENGLISH

2.1 操作性定义：三个承重条款

An Operational Definition: Three Load-Bearing Clauses

底本给出了一个包含三个承重条款的正面定义：嵌入客户约束所在的工程师；以客户真实案例构建生产级机制；并负有把部署所学转化为产品的内置职责。去掉嵌入，剩下的是供应商；去掉机制，剩下的是顾问；去掉产品职责，剩下的是人力外包。本研究采用这一定义，因为它可以在产物层面被检验。

The source offers a positive definition with three load-bearing clauses: an engineer embedded at the customer's constraint, building production machinery from the customer's real cases, with a wired-in duty to convert deployment learning into product. Each clause excludes an adjacent role. Remove the embedding and what remains is a vendor; remove the machinery and what remains is a consultant; remove the product duty and what remains is a body shop. The study adopts this definition because it can be checked at the level of artifacts.

概念界定 DEFINITION

前线部署工程师（FDE） *Forward-deployed engineer (FDE)*

嵌入客户约束所在、以客户真实案例构建生产级机制，并负有将部署所学转化为产品之职责的工程师。其不变内核是：对客户的结果负责。

An engineer embedded at the customer's constraint, building production machinery from the customer's real cases, with a duty to convert deployment learning into product. Its invariant core: accountability for the customer's outcome.

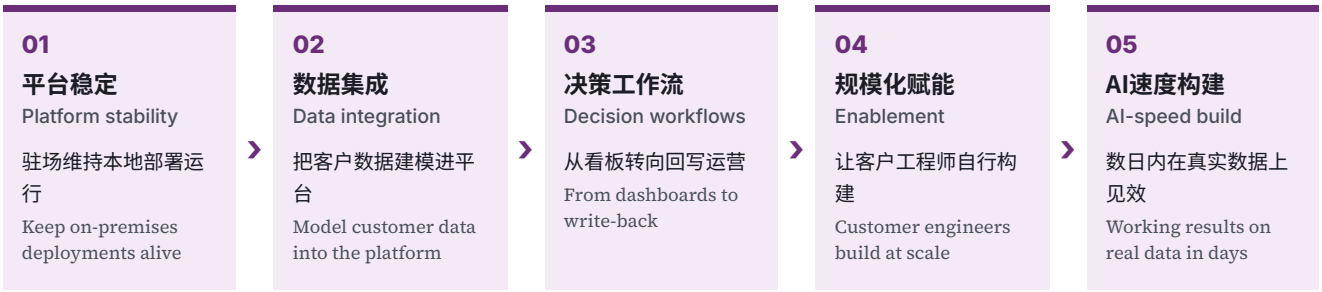
2.2 角色演进的五个年代

Five Vintages of the Role

据底本整理的从业者叙述，这一角色在同一头衔下经历了五个年代：平台稳定、数据集成、决策工作流、规模化赋能，以及当下的AI速度构建。贯穿五个年代的不变量从来不是任务清单，而是对客户结果的问责。

Practitioner accounts compiled in the source describe the role moving through five vintages under one title: platform stability, data integration, decision workflows, enablement at scale, and the present AI-speed build. The constant across all five was never a task list but accountability to the customer's outcome.

图 2-1 FDE角色的五个年代
Figure 2-1 *Five vintages of the FDE role*



来源 Source: Cuofano (2026), Ch. II & XX，据从业者公开叙述 / from practitioners' public accounts.

2.3 三层分析框架
A Three-Level Analytical Framework

本研究把底本的材料组织为三层框架。在问题层，"三重停滞"解释企业AI为何失败。在机制层，五个顺序机制描述项目如何攻克这些停滞。在治理层，两个平衡力量——买方的清晰产权与供应商的"第二忠诚"——决定由谁获得复利价值。第四至七章按此结构展开。

The study organizes the source's material into a three-level framework. At the level of the problem, the three stalls explain why enterprise AI fails. At the level of the motion, five sequential mechanisms describe how an engagement attacks them. At the level of governance, two counterweights, clear title for the buyer and the second loyalty for the vendor, decide who captures the compounding value. Chapters 4 to 7 follow this structure.

图 2-2 本研究的三层分析框架
Figure 2-2 *The study's three-level analytical framework*

治理层 Governance	清晰产权与第二忠诚：决定复利价值归属于买方还是供应商	Clear title and the second loyalty decide whether buyer or vendor keeps the compounding
机制层 Motion	楔子、嵌入、构建、级联、残留物产品化五个顺序机制	Five sequential mechanisms: wedge, embedding, build, cascade, productized residue
问题层 Problem	数据、采用结构、信任三重停滞，横亘于承诺与价值之间	Three stalls of data, adoption structure and trust between commitment and value
基础层 Substrate	可通过API租用、日益趋同的通用前沿模型	General frontier models, converging and rentable by API

来源 Source: 本研究据 Cuofano (2026) 构建 / constructed by this study from the source.

问题层：三重停滞 *The Problem Level: Three Stalls*

第一重停滞是数据：以AI问题开始的项目几周内就会变成数据问题。第二重是采用结构：人手一个助手只带来不会复利的个人杠杆，职能层面的杠杆必须逐个工作流地构建。第三重是信任，而且它有时钟：决定胜负的是"见效时间"，即真实案例上的真实工作可见地成功所需的天数；错过窗口，试点便陷入炼狱。

The first stall is data: engagements that begin as AI problems become data problems within weeks. The second is adoption structure: an assistant for every employee yields individual leverage that does not compound, while leverage for a whole function must be built workflow by workflow. The third is trust, and it has a clock: the deciding variable is time-to-magic, the days until real work on real cases visibly succeeds; past that window, pilots enter purgatory.

第 3 章 · CHAPTER 3

03 研究设计与方法

Research Design and Method

本研究是对底本的结构化分析性阅读，而非实地调查。本章说明资料、分析单元与四项工作假说。

This study is a structured analytical reading of the source volume, not a field survey. This chapter sets out the corpus, the units of analysis and four working hypotheses.

中文

本研究以底本为主要语料：一部共七个部分、三十章、三篇附录和三十五幅图版的实务手册，出版于2026年8月。我们对文本进行分析性而非描述性的阅读：对每一章编码其提出的机制、面向的行动者（供应商、买方或个人）、作为证据的产物，以及被标为可证伪的主张。带日期的测量值一律视为底本2026年第三季度的读数；底本中唯一完整量化的项目——附录C的保险公司——明确是一个综合案例，本研究将其作为示例而非经验观测来分析。

ENGLISH

The study takes the source volume as its primary corpus: a practitioner playbook in seven parts, with thirty chapters, three appendices and thirty-five plates, published in August 2026. It reads the text analytically rather than descriptively, coding each chapter for the mechanism it proposes, the actor it addresses (vendor, buyer or individual), the artifact it treats as evidence, and the claims it marks as falsifiable. Dated measurements are taken as the source's Q3 2026 readings. The one fully quantified engagement in the source, the insurer of Appendix C, is explicitly a composite and is analysed as a worked illustration, not as an empirical observation.

30	35	3	4
章 chapters	**幅图版** plates	**篇附录** appendices	**项可证伪模式** falsifiable patterns

来源 Source: Cuofano (2026), First Edition, August 2026 — 语料构成 corpus composition.

3.1 分析单元 Units of Analysis

- **部署项目**：从评分卡到移交的完整弧线，是机制分析的基本单元。
- **实践**：运行FDE模式的组织，无论是供应商还是企业内部团队。
- **合同**：所有权、定价与移交条款，是价值分配的载体。
- **个人**：角色的能力栈、职业阶梯与进出路径。

由于这一模式涉及三方——运行它的企业、接受它的企业以及坐在这个位置上的工程师——本研究至少从两方视角为每一个机制编码。例如，楔子评分卡对供应商是销售工具，对买方是治理产物，对个人则是职业履历中的证据。这种多方阅读，是本研究相对于一部以运营者为读者的实务手册的主要方法差异。

- **The engagement**: the full arc from scorecard to handover, the basic unit for analysing mechanisms.
- **The practice**: the organization running the motion, whether a vendor or an internal team.
- **The contract**: ownership, pricing and handover clauses, the carrier of value distribution.
- **The individual**: the role's skill stack, career ladder, and paths in and out.

Because the motion has three parties, the firm running it, the enterprise receiving it and the engineer in the seat, the study codes every mechanism from at least two sides. A wedge scorecard, for example, is a sales instrument for the vendor, a governance artifact for the buyer, and a career exhibit for the individual. This multi-party reading is the study's main methodological

departure from a playbook, which by design addresses the operator.

3.2 工作假说
Working Hypotheses

底本附录A声明了四项承重模式，以及会使其失效的观察。本研究将其采纳为工作假说，并在发现章节中逐一评估其机制论证的强弱与现有证据的性质。

The source's Appendix A declares four load-bearing patterns, together with the observations that would retire them. The study adopts them as working hypotheses and, in the findings chapters, assesses the strength of each mechanism and the nature of the evidence offered for it.

表 3-1 四项工作假说及其证伪条件
Table 3-1 *Four working hypotheses and their falsifiers*

假说 Hypothesis	主张 Claim	证伪观察 Retiring observation
H1 楔子关卡 H1 Wedge gates	通过全部关卡的楔子更常存续并进入生产 Wedges passing all gates more often survive into production	未通过关卡的楔子存续率相近 Gate-failing wedges survive at similar rates
H2 公开仪表 H2 Published gauges	每周公开的仪表读数驱动级联扩张 Weekly published gauges drive cascade expansion	扩张与仪表是否公开无关 Expansion unrelated to gauge publication
H3 清晰产权 H3 Clear title	清晰产权提高续约率与推荐率 Clear title raises renewal and reference rates	被锁定客户续约与推荐更多 Captured customers renew and refer more
H4 收获率 H4 Harvest rate	产品化率预测毛利向软件水平移动 Productization rate predicts margins moving toward software grade	产品化率与毛利轨迹无关 Productization unrelated to margin trajectory

来源 Source: Cuofano (2026), Appendix A；表述经本研究整理 / wording arranged by this study.

方法说明 METHOD NOTE

证据分层 *Evidence tiers*

底本采用三层证据：由机制论证的时代性论断、可替换的周期性证据，以及在附录B中标注日期的读数。本研究沿用这一分层：机制按其论证评估，数字仅视为2026年第三季度的快照，非示意即注明来源。

The source uses three evidence tiers: era-level claims argued from mechanism, replaceable cycle evidence, and readings dated in Appendix B. This study keeps the tiers: mechanisms are judged on their argument, and numbers are treated as Q3 2026 snapshots, either sourced or marked illustrative.

第 4 章 · CHAPTER 4

04 发现一：楔子选择与信任时钟

Finding 1: Wedge Selection and the Trust Clock

第一个发现是：部署的成败在启动前就已大体在纸面上决定。楔子选择实质上是对"信任时钟"这一稀缺资源的预算管理。

中文

The first finding is that a deployment is largely won or lost on paper before it starts. Choosing the wedge is, in substance, budgeting a scarce resource: the trust clock.

ENGLISH

4.1 四道关卡与两个过滤器

Four Gates and Two Filters

底本的第一个操作性主张是：每次部署都始于一个决定后续大部分走向的选择——先解决哪个问题。楔子是最快转化为可见、可验证生产价值的问题，因为首个胜利会换来此后一切的授权。候选问题须通过数量、可规定性、可验证性与痛感四道关卡；另有可逆性与数据邻近两个过滤器保护时钟。

The source's first operational claim is that every deployment begins with a choice that decides most of what follows: which problem to solve first. The wedge is the problem that converts fastest into visible, verifiable production value, because the first win buys the mandate for everything after it. A candidate must pass four gates, volume, specifiability, verifiability and pain, while two filters, reversibility and data proximity, protect the clock.

表 4-1 楔子评分卡的六项标准——以保险公司代位追偿审核为例
Table 4-1 *The six wedge criteria, applied to the insurer's subrogation review*

标准 Criterion	检验问题 Test question	附录C中的读数 Appendix C reading
数量 Volume	能否在一个季度内收回构建成本? Will the unit earn its build within a quarter?	每年约11,000件 About 11,000 cases a year
可规定性 Specifiability	能否写下正确结果? Can correct outcomes be written down?	有保单依据 Policy-specifiable
可验证性 Verifiability	能否用真实案例核验? Can results be checked on real cases?	可对照已裁决的历史 Checkable against adjudicated history
痛感 Pain	是否有高管每周都感受到? Does someone senior feel it weekly?	一位具名副总裁每周感受 Felt weekly by a named VP
可逆性 Reversibility	是否有天然的退出点? Is there a natural escape point?	位于支付关口之前 Behind the payment gate
数据邻近 Data proximity	数据能否在数日内取得? Is the data reachable in days?	数日内可得 Reachable in days

来源 Source: Cuofano (2026), Ch. VII & Appendix C（综合案例 composite case）.

4.2 三种反模式
Three Anti-Patterns

关卡同样由它所排除的东西来定义。登月式楔子无法通过可规定性关卡，并在探索中耗尽信任窗口；琐碎楔子通过所有技术关卡却依然失败，因为无人在意的胜利招不来第二份章程；演示楔子把供应商的便利误当作客户的约束，差距在第三周以数据问题暴露。关卡是一种政治工具，而非技术规范。

The gates are defined as much by what they exclude. The moonshot wedge fails specifiability and burns the trust window on discovery. The trivial wedge passes every technical gate and still loses, because a win nobody feels recruits no second charter. The demo wedge confuses the vendor's convenience with the customer's constraint, and the gap surfaces in week three as a data problem. The gates are a political instrument, not technical hygiene.

图 4-1 楔子选择矩阵
Figure 4-1 *The wedge selection matrix*

↑ **业务痛感** Felt pain

登月式楔子 Moonshot wedge 痛感强却无法规定与验证，在探索中耗尽信任窗口 Felt but unspecifiable; burns the trust window on discovery	**真正的楔子** True wedge 高痛感且可核验：首个胜利换取后续授权 Felt and verifiable; the first win buys the mandate
演示楔子 Demo wedge 迁就供应商演示，未在客户案例上验证 Fits the vendor's demo, unproven on the customer's cases	**琐碎楔子** Trivial wedge 技术关卡全过，却无人在意，招不来第二份章程 Passes technical gates; nobody feels it, no second charter

可验证性 Verifiability →

来源 Source: 本研究据 Cuofano (2026) 第七章归纳 / derived from Ch. VII.

4.3 信任时钟与见效时间
The Trust Clock and Time-to-Magic

楔子之所以关键，源于第三重停滞：决定结局的是"见效时间"，即客户的真实案例在生产条件下首次可见成功的天数，业界目标是"以周计，而非以季度计"。楔子评分卡因而是管理组织耐心这一稀缺资源的预算工具。

The wedge matters because of the third stall: the deciding variable is time-to-magic, the days until the customer's real cases visibly succeed in production, with a field target of weeks, not quarters. The wedge scorecard is thus a budgeting tool for a scarce resource: organizational patience.

研究发现 FINDING

发现一 *Finding 1*

楔子选择是治理决策而非技术决策。评分卡用客户自己的数字作出选择，并作为客户拥有的第一件产物交付，从而把高层偏好转化为可审计的依据；登月项目应以书面形式、以评分卡为理由予以拒绝。

Wedge selection is a governance decision, not a technical one. The scorecard decides on the customer's own numbers and is handed over as the first artifact the customer owns, turning executive preference into an auditable rationale; moonshots are declined in writing, with the scorecard as the reason.

第 5 章 · CHAPTER 5

05 发现二：嵌入、构建与智能体节奏

Finding 2: Embedding, Building and Agent Cadence

第二个发现是：嵌入并非座位安排，而是萃取"特定性"的方法；构建的产物不仅是机制，更是客户基于证据建立起来的信心。

The second finding is that embedding is a method for extracting the specific, not a seating arrangement, and that the build's product is not only machinery but the customer's evidence-based confidence in it.

中文

ENGLISH

5.1 嵌入的三项纪律

Three Disciplines of Embedding

按底本的叙述，嵌入的第一周是"考古"。文档中的流程与实际执行的流程处处是两回事，部署往往死在二者的缝隙里。FDE与真正做事的人而非其经理坐在一起，记录例外率、变通做法、未写入规程的判断，以及一切暗中依赖的那张电子表格，并自第一天起写入客户系统上、以客户名义保存的记录。

In the source's account, the first embedded week is archaeology. The documented and the practised process are different objects everywhere, and deployments die in the gap. The FDE sits with the people who do the work rather than their managers, recording exception rates, workarounds, unwritten judgment calls and the spreadsheet everything secretly runs on, and from day one files it into a record on the customer's systems, in the customer's name.

第二周是数据底座：开通权限、收集案例，汇集六十至一百个真实且已裁决的案例构成的黄金集。在这里，AI问题显露为数据问题，对策是"范围诚实"：只构建楔子所需的最小底座，把更大的数据债务记为发现。第三项纪律是"在场"：约束的未成文部分只能在现场听到，远程交付恰恰失去这一通道——这一模式遭诟病的高成本，正是它真正的机制。

The second week is substrate: access provisioned, cases collected, and a golden set of sixty to one hundred real, adjudicated examples assembled. Here the AI problem reveals itself as a data problem, and the response is scoped honesty: build the minimum substrate the wedge needs and log the larger data debt as findings. The third discipline is presence: the unwritten part of the constraint is audible only on the floor, and remote delivery loses that channel, so the motion's derided expense is its actual mechanism.

5.2 现场构建的三处顺序调整

Three Changes to the Order of Operations

底本认为，现场条件在三个方面改变了一套原本标准的工程方法的顺序。第一，评测集先于单元：冻结的评测集同时是政治工具，是日后向赞助人和怀疑者证明胜利的中立裁判；由于客户的专家签署正确答案，它不会被视为供应商自评。第二，框架用客户材料构建：客户的记录、合规规则、语气与阈值，沿着"文档—审阅智能体—机械检查"的层级被逐级提升。第三，单元公开毕业，其自主性阈值在首次运行前就与赞助人商定。

The source argues that field conditions change the order of an otherwise standard engineering method in three ways. First, the suite comes before the unit: the frozen evaluation suite is also a political instrument, the neutral referee that will later prove the win to sponsors and skeptics, and because the customer's expert signs the correct answers it cannot be dismissed as self-grading. Second, the harness is built from customer material: the customer's record, compliance rules, tone and thresholds, promoted from documentation to reviewer agents to mechanical checks. Third, the unit graduates in public, against

autonomy thresholds agreed with the sponsor before the first run.

由此得出本章的核心洞见：现场构建的产物不只是机制，还有客户对机制"挣来的"信心，而信心由按时交付的证据建立，从不由保证建立。硬性规则随之而来：一切都在客户的"车间"里运行，因为改进循环运行在供应商平台内的单元，是在为"房东"积累复利。

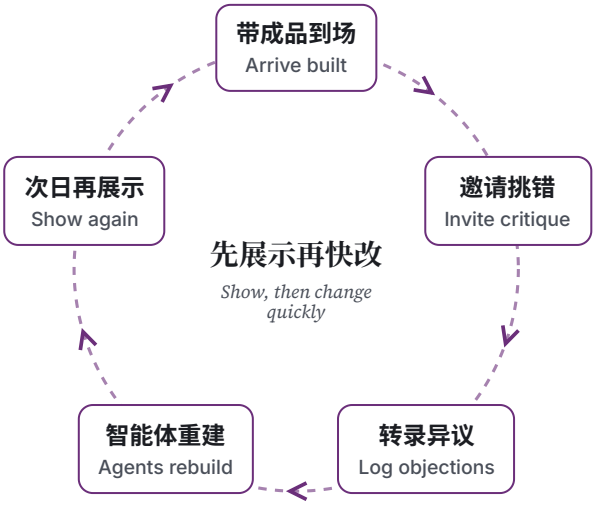
Hence the chapter's central insight: the product of a field build is not only the machinery but the customer's earned confidence in it, and confidence is built by evidence delivered on schedule, never by assurance. A hard rule follows: everything runs on the customer's floor, because a unit whose improvement loop runs inside the vendor's platform compounds for the landlord.

5.3 智能体节奏
Agent Cadence

底本记录了一种由开创者公开示范的方法：现场团队带着已可运行的版本到场，开场即邀请客户指出错在哪里；上午的异议被转录并交给智能体，下午完成重建，次日再循环。三四天的往复即可产出过去需数月的成果。更深层的主张关乎复利：一周迭代十次的团队与一周迭代一次的团队，到月底处于不同的状态。

The source documents a method demonstrated publicly by the originator: the field team arrives with a working version and opens by inviting the customer to say why it is wrong; morning objections are transcribed and fed to agents, the build is redone by afternoon, and the cycle repeats next day. Three or four days of such turns produce what scoped projects took months to deliver. The deeper claim is compounding: a team iterating ten times a week is not ten percent better than one iterating once; it operates in a different regime.

图 5-1 智能体节奏：上午挑错，下午重建
Figure 5-1 *Agent cadence: critique by morning, rebuild by afternoon*



来源 Source: Cuofano (2026), Ch. IX.

60–100
黄金集案例数
golden-set cases

3–4天
一轮驻场构建
per on-site build

10×
同周迭代次数之比
iterations in the same week

来源 Source: Cuofano (2026), Ch. VIII–IX；"10×"为底本的对比性表述，并非测量值 / '10×' is the source's comparative framing, not a measurement.

第 6 章 · CHAPTER 6

06 发现三：级联、残留物与复利经济

Finding 3: Cascade, Residue and Compounding Economics

第三个发现是：FDE模式的经济性不在单个项目，而在两次转化——单元到项目群的级联，以及跨客户重复部分到产品的"收获"。

中文

The third finding is that the motion's economics lie not in single engagements but in two conversions: the cascade from unit to programme, and the harvest of what repeats across customers into product.

ENGLISH

6.1 从项目到项目群

From Project to Programme

一个经过验证的单元只是一个项目；级联把它变成项目群，而按底本的说法，级联靠数字而非热情运转。当首个单元的九十天面板显示人工关注得到释放、产出上升、单位结果成本下降——且是在客户自己冻结的评测集和案例上测得——相邻团队会在楔子的形状中看到自己的工作，并要求属于自己的版本。正确的回应是有纪律的：每个候选都公开地走同一张评分卡，使扩张继承而非消耗首胜的信誉。这正是H2背后的机制；保险公司综合案例中，两个相邻团队提交候选，其中一个通过。

One proven unit is a project; the cascade turns it into a programme, and in the source's account it runs on numbers, not enthusiasm. When the first unit's ninety-day panel shows attention returned, yield rising and cost per outcome falling, measured on the customer's own frozen suite and cases, adjacent teams recognise their work in the wedge's shape and ask for their version. The disciplined response is to run every candidate through the same scorecard in the open, so that expansion inherits the first win's credibility instead of spending it. This is the mechanism behind H2; in the insurer composite, two adjacent teams arrived with candidates and one passed.

级联内部存在一次结构转换。第一个单元是"零售"的：一个工作流，手工打造。到第三、第四个单元时，"批发"层浮现：每个单元都需要的共享记录、权限管道、仪表基础设施与评测工具。底本认为，真正的"从单人到多人"转变发生在这一使下一个单元更便宜的共享机制中，而非培训推广中；随之而来的治理时刻，则由从第一个单元起就维护的单元名册来回答。

Inside the cascade sits a structural shift. The first unit is retail: one workflow, hand-built. By the third or fourth unit the wholesale layer appears: the shared record, permission plumbing, gauge infrastructure and suite tooling every unit needs. The source locates the real single-player-to-multiplayer transition in this shared machinery, which makes each next unit cheaper, not in training rollouts; and the governance moment that follows is answered by a roster of units kept from unit one.

6.2 残留物的收获层级

The Harvest Hierarchy of Residue

"第二忠诚"基于一个简单的观察把部署转化为生意：客户足够多时，特定的东西会重复出现。每次部署都会留下残留物——三个客户都需要的录入模式、每个受监管行业都要的审阅视角、重建了四次的仪表板、连接人人都在用的系统的接口。底本要求建立一条从现场通往产品的常设通道，并在每次复盘时追问：我们在这里第二次构建了什么？收获随后遵循一个层级：模式先被记录为操作手册，

The second loyalty converts deployments into a business on a simple observation: across enough customers, the specific repeats. Every deployment leaves residue, such as an intake pattern three customers needed, a reviewer lens every regulated industry asks for, a dashboard rebuilt four times, or a connector to a system everyone runs. The source prescribes a standing channel from field to product and a standing question at every review: what did we build

承重的模式被提升为工具包，经过验证的模式再被提升为产品。每一次提升都以不同方式改变利润率。

here for the second time? Harvest then follows a hierarchy. Patterns are captured as playbook, load-bearing patterns are promoted to toolkit, and proven patterns are promoted to product. Each promotion moves margin in a different way.

图 6-1 残留物的收获层级
Figure 6-1 *The residue harvest hierarchy*

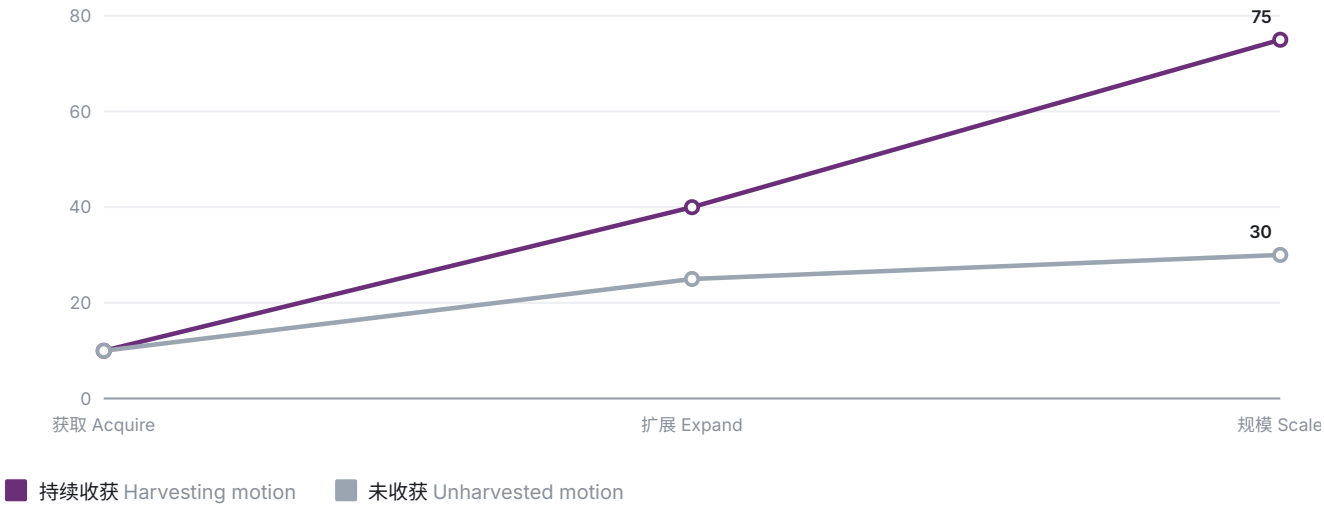
产品 Product	平台功能，按软件定价、数日部署：改变商业模式	Platform features priced as software, deployed in days; changes the business model
工具包 Toolkit	现场团队随身携带的脚手架、模板与检查：缩短时间	Scaffolds, templates and checks the field team ships with; cuts time
操作手册 Playbook	下一位FDE可直接执行的书面方法：降低成本	Written method the next FDE runs; cuts cost
部署残留 Residue	每次部署留下的、跨客户重复出现的模式	Patterns each deployment leaves that repeat across customers

来源 Source: Cuofano (2026), Ch. XI.

底本把开创者的上市文件视为这一模式的奠基文本。文件把客户关系分为三个阶段：获取，常以低利润甚至负利润进行，由现场团队承担了解机构的学习成本；扩展，随早期部署得到验证而深化；规模化，现场投入逐步退出，关系转化为高利润的软件收入。贡献利润被设计为逐阶段攀升——现场工作被组织为一项有明确回收期的投资。底本进一步主张，推动这条曲线的正是收获：企业利润率与软件基准之间的差距，就是它尚未收获的残留物。

The source treats the originator's public listing documents as the motion's founding text. They describe customer relationships in three phases: acquire, often at low or negative margin while the field team absorbs the cost of learning the institution; expand, as early deployments prove out; and scale, as field investment falls away and the relationship converts to high-margin software. Contribution margins were shown climbing through the phases by design, so that field work was structured as an investment with a documented payback. The source's further claim is that harvest moves the curve: the gap between a firm's margin and the software benchmark is its unharvested residue.

图 6-2 获取—扩展—规模阶段的贡献利润路径（示意）
Figure 6-2 *Contribution margin across acquire, expand and scale (illustrative)*



示意 Illustrative: 指数为假设值，仅表现底本描述的曲线形状（Ch. II, XI, XXIII），非实际数据 / hypothetical index showing the shape the source describes, not data.

6.3 六个仪表
The Six Meters

为区分复利型模式与出租工程师的模式，底本提出六个须合并、按趋势阅读的仪表：**见效时间**、**扩张率**、**结果转化率**（同时审计构建纪律）、**产品化率**（预测利润曲线）、**毛利轨迹**与**产权清洁度**。旁边还有一份拒绝日志，记录被拒绝的项目及原因。

To tell a compounding motion from a rented one, the source proposes six meters read together and on trend: **time-to-magic**, **expansion rate**, **outcome conversion** (which also audits build discipline), **productization rate** (which predicts the margin curve), **gross margin trajectory** and **title cleanliness**. Beside them sits a refusal log of engagements declined and why.

研究发现 FINDING

发现三 *Finding 3*

级联与收获是同一复利机制的两面：级联在客户内部把零售单元沉淀为批发底座，收获在客户之间把"特定"沉淀为"通用"。二者都只有在被排期保护时才会发生——满负荷计费的团队产生收入，却不产生残留物。

Cascade and harvest are two faces of one compounding mechanism: the cascade condenses retail units into wholesale substrate within a customer, and the harvest condenses the specific into the general across customers. Both happen only when scheduled: a team billed at full utilisation produces revenue but no residue.

残留物命题最终指向FDE自身的工作：考古与评测集构建本身正被编码进产品，这份工作就是把这份工作自动化。对H4而言，这是书中最尖锐的预测：产品化率应领先于毛利轨迹。

The residue thesis finally turns on the FDE's own job: archaeology and suite construction are themselves being encoded into product, so the job is to automate the job. For H4 this is the volume's sharpest prediction: productization rate should lead margin trajectory.

1/季	2	1
首年每季残留物提升 residue promotion a quarter	**资深FDE可承担中型项目** mid-sized engagements per senior	**首年凭公开仪表扩张** account expanded on gauges

来源 Source: Cuofano (2026), Ch. XXI — 底本建议的首年预期，非观测结果 / first-year expectations recommended by the source, not observed results.

第 7 章 · CHAPTER 7

07 发现四：控制权问题与诚实合同

Finding 4: The Capture Question and the Honest Contract

第四个发现是：嵌入式工程师既是最快的能力转移通道，也是最深的依赖机制；二者之间的差别由合同结构决定，而不由善意决定。

The fourth finding is that the embedded engineer is both the fastest channel of capability transfer and the deepest dependency mechanism, and that the difference is decided by contract structure, not by good intentions.

中文

ENGLISH

7.1 一个双刃机制

A Double-Edged Mechanism

底本毫不含糊地陈述了买方的两难：一位为你构建运营机制的嵌入式工程师，既是这个时代最快的能力转移，也是有史以来最深的依赖机制。当离开一家AI供应商的成本被逐项定价时，最大的几项恰恰是FDE项目的产物：评测集、编码化的标准、积累的记录与调校过的工作流。如果它们建在供应商平台内、使用其格式、只对其系统可读，那么项目每推进一周，客户就在为加深自身依赖付费。这并非恶意，而是非结构化嵌入的默认结果。

The source states the buyer's dilemma without euphemism: an embedded engineer building your operational machinery is the fastest capability transfer of the era and the deepest dependency mechanism ever sold. When the cost of leaving an AI vendor is priced line by line, the largest lines are exactly what an FDE engagement produces: evaluation suites, encoded standards, accumulated records and tuned workflows. If these are built inside the vendor's platform, in its formats and legible only to its systems, every week of the engagement deepens a dependency the customer is paying to acquire. This is not bad faith but the default outcome of unstructured embedding.

买方已开始以“机构主权”命名这一关切：企业对其业务决策权的所有权。它包含四种担忧——锁定、向潜在竞争者支付的补贴、第三方未经同意的规则变更，以及为自有专用模型即可完成的工作支付前沿价格。主权是一个从合同保护到自有模型的梯度，按用例选择；其最深的资产是决策循环本身：所见数据、所用推理、所采行动、所观结果。

The buyer side has begun naming its concern as institutional sovereignty: the enterprise's ownership of its business's decision rights. It carries four worries: lock-in, subsidy paid to a potential competitor, rule changes made without consent, and frontier prices for work an owned task-specific model could do. Sovereignty is a gradient from contractual protection to owned models, chosen per use case, and its deepest asset is the decision loop itself: data seen, reasoning applied, action taken, outcome observed.

表 7-1 签约前的五个产权问题
Table 7-1 *The five clear-title questions before signature*

对象 Object	检验问题 Test question	被俘获的信号 Capture signal
评测集 Suite	明天能否在另一家供应商的模型上运行? Can it run on another vendor's model tomorrow?	只能在供应商平台内运行 Runs only inside the vendor platform
记录 Record	存放在谁的系统、采用谁的格式? On whose systems, in whose formats?	专有格式，无法导出 Proprietary, not exportable
编码标准 Standards	审阅提示与机械检查归谁所有? Who owns reviewer prompts and checks?	以平台功能的形式存在 Exist only as platform features
模型接口 Model joint	单元能否经受模型替换? Can units survive a model swap?	模型贯穿编织于各处 Model woven through
运营者 Operators	移交后由谁运营? Who operates after handover?	需要供应商无限期驻场 Vendor presence needed indefinitely

来源 Source: Cuofano (2026), Ch. XII；"被俘获的信号"一栏为本研究归纳 / the capture-signal column is this study's synthesis.

7.2 诚实合同的结构
The Structure of the Honest Contract

底本给出的对策是一份在项目开始前写成的合同。其脊梁是所有权附表——一张表格而非一段文字——列出项目将产生的每一类产物（评测集、记录、标准、提示、检查、仪表板、连接器、手册），并为每一类指定所有者与可移植性要求。清晰的默认规则是：客户拥有以其实质内容构建的一切，供应商拥有通用模式与既有工具包，模糊之处须在第一周前逐项协商。可移植性须可检验：评测集能在替代模型上运行，记录能导出为开放格式，并在每个里程碑验证。对称的"模式许可"则允许供应商保留解决方案的匿名化形状。

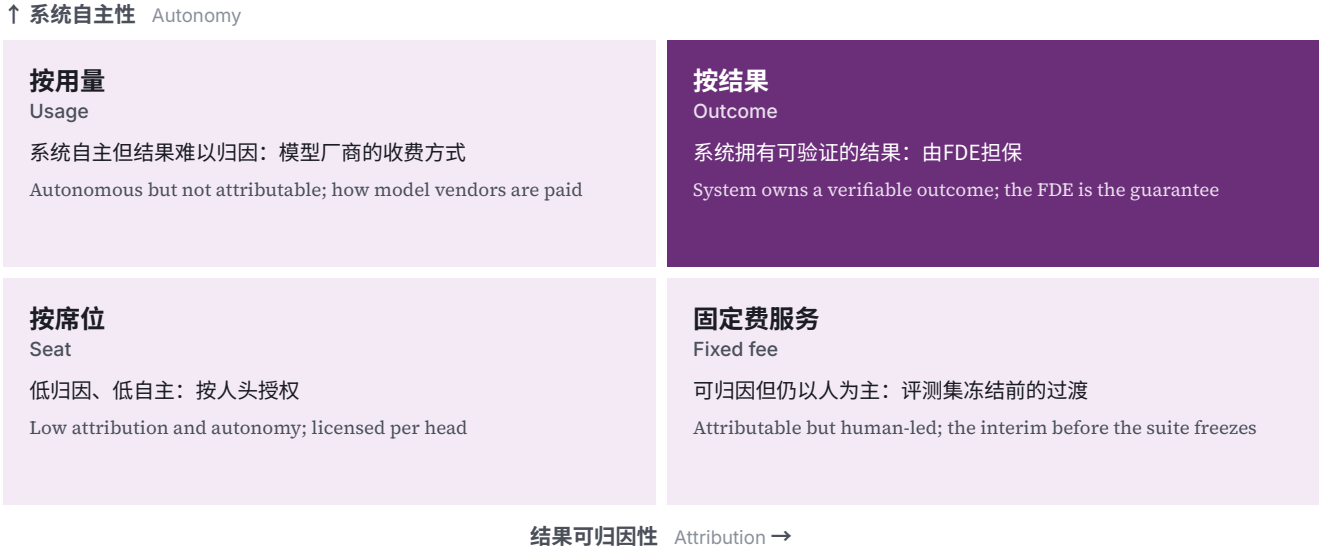
The source's remedy is a contract written before the engagement begins. Its spine is an ownership schedule, a table rather than a paragraph, listing every artifact class the engagement will produce (suites, records, standards, prompts, checks, dashboards, connectors, playbooks) with an owner and a portability requirement for each. The clean default is that the customer owns everything built from its substance, the vendor owns general patterns and pre-existing toolkit, and anything ambiguous is negotiated by name before week one. Portability must be testable: the suite runs against an alternative model and the record exports to open formats, verified at each milestone. A symmetric pattern license lets the vendor keep the anonymized shape of the solution.

定价阶梯 *The Pricing Ladder*

在底本的框架中，定价沿两个轴展开：结果在多大程度上可归因于软件，以及系统在达成结果时拥有多少自主性。低归因按席位定价；有自主性但归因不清时按用量定价，模型厂商本身就是这样收费的；当系统拥有可验证的结果时，价格就是结果。按结果定价须有人担保结果，担保者正是FDE。因此项目沿阶梯攀升：楔子与首个单元按固定费用提供服务；评测集冻结后转为按结果计价，因为没有验证工具的结果定价只是一份争议日程表；残留物成为产品后转为订阅。

In the source's frame, pricing follows two axes: how attributable the outcome is to the software, and how much autonomy the system has in achieving it. Low attribution is priced by the seat; autonomy without clean attribution is priced by usage, as the model vendors themselves are paid; where the system owns a verifiable outcome, the price is the outcome. Outcome pricing needs a guarantor of the outcome, and that guarantor is the FDE. The engagement therefore climbs a ladder: fixed-fee services for the wedge and first unit; outcome terms once the frozen suite exists, since outcome pricing without a verification instrument is merely a dispute schedule; and subscription once residue has become product.

图 7-1 归因 × 自主定价网格
Figure 7-1 *The attribution × autonomy pricing grid*



来源 Source: Cuofano (2026), Ch. XIII；右下象限为本研究据定价阶梯所作推断 / bottom-right quadrant is this study's inference from the pricing ladder.

最后，移交是一项带验收测试的交付物：具名的客户员工在规定期限内运营机制，FDE在场但不动手，项目在客户证明能够运营时结束。供应商抗拒这一条款，因为它终止了计费；底本的反驳是：被俘获的客户愤然流失，拥有清晰产权的客户主动续约。依赖是对来年收入的折价——这正是H3背后的机制。

The handover, finally, is a deliverable with an acceptance test: named customer staff operate the machinery for a defined period with the FDE present but hands off, and the engagement closes when the customer demonstrates operation. Vendors resist the clause because it ends the meter; the source replies that captured customers churn angrily while customers with clear title renew by choice. Dependency is a discount on next year's revenue, the mechanism behind H3.

> **一个本可以离开却选择留下的客户，才是唯一值得拥有的推荐。**
>
> *A customer who could leave and does not is the only reference worth having.*
>
> — Cuofano (2026), Ch. XXVIII

7.3 内部模式：在组织架构中回答控制权问题
The Internal Motion: Answering Capture in the Org Chart

对部分企业而言，底本对控制权问题的回答是结构性的：自建这一职能。它提出四项检验——工作流具有战略性、监管密度高、数据引力真实存在、跨业务单元存在重复量。只满足一项的企业应以清晰产权购买这一模式；四项全部满足的企业，则是在支付供应商利润去为别人积累残留物。内部模式在资金、销售、残留物与产权四方面发生反转。其公认的良好形态是一个向各业务单元部署的中心实践；

For some enterprises, the source's answer to the capture question is structural: build the function in-house. It proposes a four-part test of strategic workflows, high regulatory density, real data gravity and repeat volume across business units. An enterprise meeting one condition should buy the motion with clean title; one meeting all four is paying vendor margins to build someone else's residue. The internal motion inverts funding, selling, residue and title. Its known good shape is a central practice deploying into business units.

第 8 章 · CHAPTER 8

08 案例研究与比较分析

Case Studies and Comparative Analysis

本章检视底本中的两个案例——模式的开创企业与一个保险业综合案例——并比较五个"物种"在技术栈不同位置运行同一模式时的经济逻辑。

This chapter examines the source's two cases, the motion's originating firm and an insurance composite, and compares the economics of five species running the same motion from different positions in the stack.

中文

ENGLISH

8.1 案例一：开创企业 Palantir

Case 1: The Originator, Palantir

底本把Palantir的公开记录视为这一模式的奠基案例。2000年代，该公司试图向极为复杂的机构——先是情报机构，后是银行、制造商和政府——出售数据集成软件，却发现通用平台一到客户手中便毫无作用。它的回答是派驻拥有代码提交权限的构建者，而不是带着汇报材料的顾问；并为前线部署工程师配备了专注于运营与采用的第二种现场角色，使每次部署同时携带机制与其所需的变革。其2020年的S-1注册文件阐述了"获取—扩展—规模化"阶段模型，贡献利润被设计为逐阶段攀升，即图6-2所示意的经济逻辑；就H4而言，这一公开案例提供了方向性支持，但无法排除规模效应或客户结构等其他解释。

The source treats Palantir's public record as the motion's founding case. In the 2000s the company set out to sell data-integration software to highly complex institutions, first intelligence agencies and then banks, manufacturers and governments, and found the general platform inert on arrival. Its answer was to send builders with commit access rather than consultants with decks, and to pair the forward-deployed engineer with a second field role focused on operations and adoption, so that every deployment carried both the machinery and the change it required. Its 2020 Form S-1 set out the acquire-expand-scale phase model, with contribution margins climbing by design: the economics sketched in Figure 6-2. For H4, this public case offers directional support but cannot rule out scale effects or customer mix.

此后的两步完成了这个案例。其一是公开执行的产品化：十年部署残留物凝结为面向数据运营和软件交付的平台，过去由现场工程师手工构建的东西作为产品发布，利润率随残留物而来。其二是压缩：面对语言模型，该公司自2023年起推出为期数天的驻场训练营项目，在一周内基于客户自有数据搭建可运行的AI工作流。按底本所读的截至2026年第二季度的披露，这家现场人员密集的公司报告了软件级利润率和数十亿美元的AI相关合同价值。底本的结论是：这一模式不是通往软件公司途中的妥协，而是软件公司的制造流程。

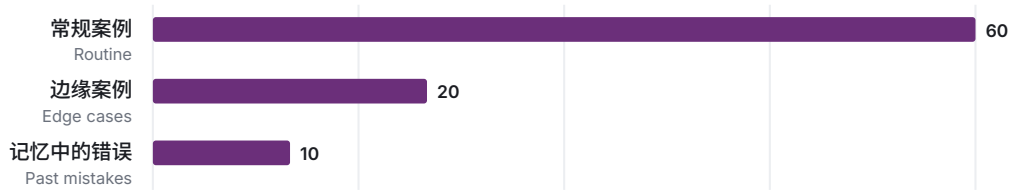
Two later moves complete the case. The first is productization executed in public: a decade of deployment residue hardened into platforms for data operations and software delivery, so that what field engineers once built by hand shipped as product, and the margins followed the residue. The second is compression: the company's response to language models was a programme of multi-day on-site bootcamps, from 2023 onward, standing up working AI workflows on customers' own data within a week. In the disclosures through Q2 2026 that the source reads, this field-heavy company reported software-class margins and multi-billion-dollar AI-attached contract value. The source's verdict: the motion was the manufacturing process of a software company, not a compromise on the way to one.

8.2 案例二：保险公司综合案例
Case 2: The Insurer Composite

附录C完整叙述了一次部署。底本明确说明这是一个综合案例——没有真实企业，但每一步都真实存在——本研究将其作为示例处理。一家中型保险公司聘请一家应用AI公司，派驻一名FDE。第零周，对赞助人清单上的五个候选问题打分；代位追偿审核胜出，而"重新构想理赔"这一登月项目以评分卡为理由被书面拒绝，评分卡由客户保留。第一、二周，FDE与四名理赔员坐在一起，汇集了九十个已裁决案例的黄金集，由保险公司的资深理赔员评分，并于第三周签字冻结。

Appendix C narrates one deployment end to end. The source states plainly that it is a composite, with no real firm but every step real, and the study treats it as a worked illustration. A mid-sized insurer engages an applied-AI firm, and one FDE deploys. In week zero, five candidate problems from the sponsor's list are scored; subrogation review wins, and the moonshot, reimagining claims, is declined in writing with the scorecard as the reason, which the customer keeps. In weeks one and two the FDE sits with four adjusters and assembles a golden set of ninety adjudicated cases, scored by the insurer's senior adjuster, whose signature freezes the suite in week three.

图 8-1 保险公司综合案例的黄金集构成（共90例）
Figure 8-1 *Golden-set composition in the insurer composite (90 cases)*



来源 Source: Cuofano (2026), Appendix C（综合案例 composite）.

第三至六周是构建。合规部门的三条红线成为附带修复说明的机械检查；双视角审阅小组编码了资深理赔员的标准；模型置于清晰接口之后，并在第五周用另一家供应商的模型运行冻结评测集加以证明。首个生产胜利出现在第二十六天。单元在第九周毕业；两个相邻团队提交候选，一个通过；那张分诊表模式已第三次出现，被提升为工具包。

Weeks three to six are the build. Three compliance red lines become mechanical checks with remediation text; a two-lens reviewer panel encodes the senior adjuster's standards; and the model sits behind a clean joint, proved in week five by running the frozen suite against a second vendor's model. The first production win lands on day twenty-six. The unit graduates in week nine; two adjacent teams bring candidates and one passes; and the triage-spreadsheet pattern, seen now at a third insurer, is promoted to toolkit.

26天	-60%	104	5/5
见效时间 time-to-magic	**单案人工关注** attention per case	**收尾时评测集案例** suite cases at close	**产权问题达标** title questions met

来源 Source: Cuofano (2026), Appendix C — 综合案例的九十天读数，非真实企业观测 / ninety-day readings of a composite, not an observed firm.

第十三周进行移交测试：两名具名员工运营三周，FDE在场但不动手；随后项目为级联部分转为按结果计价。这个综合案例与其说是证据，不如说是一致性检验：第四至七章的每个机制都在有日期的节点出现。其结尾——保险公司本可离开，却选择了扩张——正是底本对H3的浓缩表述。

Week thirteen brings the handover test: two named staff run the cadence for three weeks with the FDE present and hands off, after which the engagement converts to outcome terms for the cascade. The composite is less evidence than a consistency check: every mechanism of Chapters 4 to 7 appears at a dated point. Its closing line, that the insurer could leave and expands instead, is the source's compressed statement of H3.

8.3 比较分析：同一模式的五个物种
Comparative Analysis: Five Species of One Motion

底本把同一模式映射到技术栈的五个位置，而位置决定了部署的目的。在模型层，前沿实验室派驻FDE，把代币承诺从试用变为结构性收入。在数据平台层，模式按消耗变现，残留物沉淀为连接器和模式目录。在应用层——开创者所在的位置——部署就是产品的制造流程，以严格的产权为护城河。服务层既有沿"费用—结果—产品"阶梯攀升的AI原生公司，也有被迫转型的咨询公司。第五个物种在企业内部：买方自建。

The source maps the same motion onto five positions in the stack, and position decides what the deployment is for. At the model layer, frontier labs field FDEs to turn token commitments from provisional to structural. At the data-platform layer, the motion monetizes consumption, and residue hardens into connectors and pattern catalogues. At the application layer, the originator's position, the deployment is the product's manufacturing process, with tight title as the moat. The services layer holds AI-native firms climbing a fee-to-outcome-to-product ladder alongside retrofitting consultancies. Inside the enterprise sits the fifth species: the buyer growing its own.

图 8-2 同一模式的五个物种
Figure 8-2 *Five species of one motion*

模型层 Model layer	前沿实验室：把代币承诺从试用变为结构性收入	Frontier labs turn token commitments from provisional to structural
数据平台层 Data platform	按消耗变现；残留物沉淀为连接器与模式目录	Monetize consumption; residue becomes connectors and catalogues
应用层 Application	部署即产品制造流程，以严格产权为护城河	Deployment as manufacturing process, tight title as the moat
服务层 Services	AI原生公司的费用—结果—产品阶梯，及转型中的咨询公司	AI-native firms on a fee-outcome-product ladder; retrofitting consultancies
企业内部 Enterprise	买方自建：资金来自所服务的损益，残留物留在本企业	The buyer grows its own; the served P&L pays and residue stays home

来源 Source: Cuofano (2026), Ch. IV & XIV.

第 9 章 · CHAPTER 9

09 讨论与结论：启示、局限与研究议程

Discussion and Conclusions: Implications, Limits and a Research Agenda

本章综合四项发现，讨论其对供应商、买方与个人的启示，检视失败模式与研究局限，并提出未来研究议程。

This chapter draws the four findings together, discusses what they imply for vendors, buyers and individuals, examines failure modes and the study's limits, and proposes a research agenda.

中文

ENGLISH

9.1 对三方的启示
Implications for the Three Parties

对供应商而言，各项发现汇聚为一条设计原则：现场项目是一项投资，其回报取决于有排期的收获和清晰的产权。底本关于建立模式的建议由此而来：只在两三次由创始人主导、验证了可重复楔子形状的部署之后才启动；以一位首席级FDE和一位商业搭档作为创始组合；在扩充人手之前先装好部署记录、收获通道与仪表；并保留拒绝日志。

For vendors, the findings converge on one design principle: the field engagement is an investment whose return depends on scheduled harvest and clean title. The source's guidance on building a motion follows directly. Start only after two or three founder-led deployments have proved a repeatable wedge shape; hire a founding pair of a principal-grade FDE and a commercial counterpart; install the deployment record, the harvest channel and the meters before scaling headcount; and keep a refusal log.

对买方而言，决定性的谈判发生在演示打动人心之前：五个产权问题、所有权附表和移交验收测试，把控制权问题从信任问题变为验证问题。对个人而言，底本描述了一种"对角线"画像——足以独立构建的深度、被现场接纳的气质、能够收获的产品感——以及一条晋升依据产物而非资历的范围阶梯。

For buyers, the decisive negotiation happens before the demo impresses: the five title questions, the ownership schedule and the handover acceptance test turn the capture question from a matter of trust into one of verification. For individuals, the source describes a diagonal profile, with depth to build alone, temperament to be welcomed and product sense to harvest, and a ladder of scope on which promotion runs on artifacts, not tenure.

9.2 失败模式
Failure Families

表 9-1 六类部署失败及其结构性修复
Table 9-1 *Six failure families and their structural fixes*

失败类型 Family	症状 Symptom	结构性修复 Structural fix
试点炼狱 Pilot purgatory	无限演示，赞助人轮换后夭折 Demos forever; dies when sponsors rotate	楔子关卡；信任时钟作硬预算 Wedge gates; trust clock as hard budget
人力外包漂移 Staff-aug drift	FDE沦为工单队列中的承包商 FDE absorbed into the ticket queue	每个检查点复盘残留物 Residue review at every checkpoint
英雄依赖 Hero dependency	作者离开后单元漂移 Units drift after the author leaves	书面优先；移交验收测试 Written-first; handover acceptance test
俘获反噬 Capture backfire	新CTO做退出成本分析后丢单 Account lost to an exit-cost review	合同清晰产权；里程碑验证 Clear title by contract; milestone checks
登月楔子 Moonshot wedge	探索阶段耗尽窗口 Discovery consumes the window	评分卡不向职级让步 Scorecard enforced against seniority
未收获的模式 Unharvested motion	二十次部署，一无所有 Twenty deployments, owning nothing	收获层级与定期提升评审 Harvest hierarchy reviewed on cadence

来源 Source: Cuofano (2026), Ch. XV（综合失败案例 composite cases）.

这一分类与四项发现恰好对应。底本的对策是分为四环的"排期判断"：每周按仪表复盘项目，每月复盘残留物与记录，每季度按六个仪表和拒绝日志复盘整个实践，每年做一次"从零开始"的复盘，追问哪些现场工作到明年应当成为软件。

The taxonomy maps neatly onto the four findings. The source's remedy is scheduled judgment in four rings: weekly engagement reviews on the gauges, monthly reviews of residue and record, quarterly practice reviews on the six meters and the refusal log, and an annual clean-sheet review asking which parts of the field work should be software by next year.

9.3 研究局限与风险
Limitations and Risks

本研究的局限应予直言。语料是一部从运营者立场写成的单一实务著作，其核心机制的证据是论证性的而非测量性的——底本自己也把楔子关卡、合同结构、失败案例与仪表标为"方法"。唯一量化的项目是综合案例；公开企业证据集中于少数公司；2026年第三季度的市场读数会很快过时。四项假说均未在部署总体上得到检验。

The study's limitations should be stated plainly. Its corpus is a single practitioner volume written from the operator's side, and its evidence for the core mechanisms is argued rather than measured; the source itself labels the wedge gates, contract structures, failure composites and meters as method. The one quantified engagement is a composite, public-company evidence rests on few firms, and Q3 2026 market readings will age quickly. None of the four hypotheses is tested on a population of deployments.

- **头衔通胀**：同一头衔覆盖五种不同工作，造成研究与招聘中的测量混淆。
- **反身性自动化**：FDE的手工工作本身正在被产品化，现有机制可能迅速过时。

- **Title inflation**: one title covers five different jobs, confusing measurement in research and hiring.
- **Reflexive automation**: the FDE's manual work is itself being productized, so current mechanisms may date quickly.

9.4 结论
Conclusions

本研究的结论是：前线部署工程师最好被理解为一种具有三项耦合功能的组织机制——它在客户现场从通用中萃取特定；它通过公开、可验证的仪表，把客户挣来的信心转化为扩张；它把重复出现的部分收获为产品。这一机制究竟在构建客户的能力还是供应商的锁定，取决于合同结构而非意图。底本的浓缩表述作为理论是自洽的，但其预测仍待系统检验。

The study concludes that the forward-deployed engineer is best understood as an organizational mechanism with three coupled functions: it extracts the specific from the general on the customer's floor; it converts the customer's earned confidence into expansion through published, verifiable gauges; and it harvests what repeats into product. Whether this mechanism builds the customer's capability or the vendor's lock-in is decided by contract structure, not intent. The source's compression holds together as a theory of the role, but its predictions still await systematic test.

核心洞见 KEY INSIGHT

不会被自动化的部分 *What will not be automated*

底本预期，这一模式描述的许多工作终将被自动化；不会被自动化的，是它所编码的决策结构：从哪里进入、拥有什么、转移什么、收获什么。这些都是关于"位置"的判断。

The source expects much of the work the motion describes to be automated. What will not be is the decision structure it encodes: where to enter, what to own, what to transfer, what to harvest. These are judgments about position.

9.5 未来研究方向
Future Research Agenda

1. **回溯评分（H1）**：按六项标准对历史部署的楔子打分，比较存续率。
2. **仪表公开的面板研究（H2）**：比较每周公开与不公开仪表的项目的级联扩张。
3. **产权与续约（H3）**：比较清晰产权合同与俘获型合同客户的续约率与推荐率。
4. **收获与毛利的时间序列（H4）**：跟踪供应商产品化率与毛利轨迹的领先—滞后关系。
5. **自动化FDE工作**：研究考古与评测集构建的产品化如何重塑角色的年代。

1. **Retrospective scoring (H1)**: score historical wedges on the six criteria and compare survival rates.
2. **Gauge publication panel (H2)**: compare cascade expansion where gauges are and are not published weekly.
3. **Title and renewal (H3)**: compare renewal and reference rates under clean-title and captured contracts.
4. **Harvest and margin series (H4)**: track lead-lag between vendors' productization rate and margin trajectory.
5. **Automating FDE work**: study how productizing archaeology and suite building reshapes the role's next vintage.

> **读数会过期，工具不会。**
>
> *The readings expire; the instruments do not.*
>
> — Cuofano (2026), The Library

参考文献与研究说明 *References & Research Notes*

研究说明 *Note on Method and Sources*

本研究是对底本《前线部署工程师》（Cuofano，2026年8月第一版）的分析性阅读，而非独立的实地调查。文中数字均为底本报告的2026年第三季度读数，或明确标注为示意；附录C的保险公司案例为底本所述综合案例，不对应任何真实企业。表格与图示中的归纳和推断已在注释中注明。

This study is an analytical reading of the source volume, The Forward-Deployed Engineer (Cuofano, first edition, August 2026), not an independent field survey. Figures are the source's dated Q3 2026 readings or are explicitly marked illustrative; the insurer of Appendix C is a composite described by the source and corresponds to no real firm. Syntheses and inferences in tables and figures are flagged in their notes.

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